

4(a)	<i>answers must be in context of <u>oxygen</u></i> <i>three from:</i> 1 <u>diffusion</u> 2 movement, down a, concentration / diffusion, gradient / from high(er) to low(er) concentration ; A in terms of partial pressure 3 passive (process) ; 4 through alveolar wall / across alveolar cells / across squamous epithelial cells (of alveolus) / across pavement cells (of alveolus) ; A squamous cells 5 across, endothelium / endothelial cells / capillary wall ; A squamous, epithelium / cells (in context of capillary wall) 6 ref. to diffusion / AW, through the phospholipid bilayer of cell surface membranes / between the phospholipids of the bilayer (of squamous or endothelial cells) ;	3
4(b)(i)	65 / 66 / 67 (%)	1

Question	Answer	Marks
4(b)(ii)	1 higher partial pressures in lungs / lower partial pressures in (respiring) tissues ; A correct values / range of values, of kPa A in alveoli <i>advantages of differences (higher v lower partial pressure)</i> <i>only higher partial pressure or lower partial pressure explanation required to gain mark</i> 2 oxygen, binds to v released from, haemoglobin or oxyhaemoglobin, is formed v dissociates / AW ; 3 (so) percentage saturation of haemoglobin (with oxygen) high v low ; 4 affinity of haemoglobin for oxygen high v low ; 5 data from Fig. 4.1 to support ; 6 (body), cells / tissues, need oxygen for aerobic respiration ;	3
4(c)	<i>idea of</i> abnormal, change in the body / condition, which produces, poor / ill-, health or absence of, good health / well-being ; A disorder / illness further detail ; e.g. having an adverse effect reduces the effectiveness of functions produces, (specific) signs / symptoms lack of social, physical and mental, well-being can be infectious or non-infectious AW	2

Question	Answer	Marks
4(d)	<i>five from:</i> 1 base substitution (in gene coding for, β-globin / polypeptide) ; 2 different / altered, mRNA codon ; 3 different tRNA brings a different amino acid (to ribosome) / leads to a change in one amino acid (in the polypeptide chain) ; 4 altered, primary structure / sequence of amino acids (in β-globin) ; 5 changed, tertiary / quaternary, structure ; 6 haemoglobin / molecule, less soluble ; 7 (haemoglobin) molecules, stick together / form fibres ; 8 (haemoglobin) less able to bind oxygen / AW ; <i>points above may be qualified</i> 9 details ; ; + e.g. thymine / T, replaces, adenine / A 10 in sixth, codon / triplet or sixth amino acid in sequence is changed (non-template strand) GTG instead of GAG / GTA instead of GAA or (template strand) CAC instead of CTC / CAT instead of CTT mRNA codon is, GUG instead of GAG / GUA instead of GAA 11 (amino acid change is) valine instead of glutamic acid ; 12 amino acid with non-polar side chain instead of polar side chain ; 13 position of amino acid, is to the exterior / faces cytosol ;	5